Proceedings of the 40th Chinese Control Conference
July 26-28, 2021, Shanghai, China

An Improved ICCP Gravity Matching Algorithm Based on Mahalanobis Distance

Yu Wang, Zhihong Deng, Wenzhe Zhang, Shengwu Zhao

Integrated Navigation and Intelligent Navigation Laboratory, Beijing Institute of Technology, Beijing 100081
E-mail: bitwangyu@qq.com

Abstract: Consider the problems of low accuracy and poor real-time performance in traditional ICCP gravity matching algorithm, this paper proposes an improved ICCP gravity matching algorithm based on Mahalanobis distance. Mahalanobis distance, instead of Euclidean distance, is applied to improve the objective function of rigid transformation iteration in traditional ICCP algorithm, which can eliminate the interference of variable correlation. At the same time, a new searching strategy based on strip search area is proposed. With the current sampling point as the center, we use the gravimeter measurement error variance to determine the width of the strip region along the normal direction of the contour, and use the Mahalanobis distance objective function value obtained in the iterative process as the length of the strip area along the direction of the contour. Therefore, this searching strategy can gradually narrow the search range of the closest contour point in the ICCP algorithm, and reduce the calculation amount of the algorithm. The simulation results show that: Compared with the traditional ICCP algorithm, the mean value of the position error of the improved algorithm is reduced by 26.89%, and the standard deviation is reduced by 58.53%, which greatly improves the accuracy of the gravity matching algorithm. And the average running time of the improved algorithm is also 42.1% shorter than the traditional ICCP algorithm, so the real-time performance is improved. The improved ICCP gravity matching algorithm based on Mahalanobis distance reduces the running time of the algorithm, improves real-time performance and positioning accuracy, and further improves the practicability of gravity matching for underwater vehicles.

Key Words: ICCP matching algorithm; Mahalanobis distance; Strip-search area

1 Introduction

During the long-term continuous operation of submarines and other large submersibles under water, if the inertial navigation system is used solely for navigation, its errors will diverge over time, which will seriously affect the accuracy of navigation and positioning. The ocean gravity field has characteristic stability and position correlation, laying the foundation for assisting long-endurance inertial navigation systems [1-3]. Due to its autonomy and stability, ocean gravity field matching positioning technology has gradually become the focus of research.

Gravity matching is the focus research on gravity assisted navigation [4]. In the 1960s, with the development of the relevant extreme value theory, the composition of the relevant extreme value system, the mathematical model of the geophysical field, and the principle of the matching algorithm were systematically summarized. Since then, the matching algorithm has developed rapidly, and successively appeared terrain contour matching algorithm, Sandia inertial-terrain assisted navigation algorithm, contour-based iterative closest point algorithm and pattern recognition neural network-based matching algorithm.

Iterative Closest Contour Point algorithm is a type of matching algorithm that has been used more in gravity matching positioning in recent years [5-8]. In 1999, Behzad K.P first introduced the ICP algorithm commonly used in the field of image registration to the underwater navigation, and therefore proposed a specific implementation form of the ICCP algorithm based on contours [10]. However, the ICCP algorithm still has many shortcomings: The ICCP algorithm is a sequence iterative matching algorithm, which means that only when sufficient trajectory points are sampled can it start matching, so the real-time performance of the algorithm is poor. At the same time, since the ICCP algorithm searches for the closest point on the counter in the entire gravity anomaly background map, this process has a large amount of calculation and high computational complexity. In addition, the ICCP algorithm assumes that the gravity measurement error is small, and the true track point must be located on the corresponding contour of the gravity measurement value. In practical applications, this assumption is not necessarily satisfied. Therefore, improving the traditional ICCP gravity matching algorithm plays a vital role in improving the positioning accuracy of gravity matching.

This paper mainly focuses on the affine invariance of Mahalanobis distance to solve the limitations of traditional ICCP algorithm's low matching positioning accuracy and poor real-time performance. An improved algorithm is proposed which bases on the Mahalanobis distance to improve the objective function. At the same time, an improved strip-searching area is proposed to gradually narrow the search range of the closest counter point in the algorithm. Finally, this paper has carried out simulation verification, which proves the rationality and effectiveness of the improved algorithm.

2 Traditional ICCP Algorithm

The ICCP algorithm is developed from the iterative closest point ICP algorithm in image registration field. Under the assumption that the distance between the target point set and the template point set is not large, the distance between the two is reduced by repeated rigid transformations (translation and rotation). The basic principle of ICCP gravity matching algorithm is shown in Figure 1. In the

*This work is supported by State Key Laboratory Foundation of China under Grant 6142218200309.

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on July 29,2024 at 06:58:43 UTC from IEEE Xplore. Restrictions apply.

picture $p_i(i = 1,2,\cdots,N)$ indicates the output track point of the inertial navigation system. $C_i(i = 1,2,\cdots,N)$ indicates the contour of gravity anomaly, which is provided by the gravity reference map known in advance. $x_i(i = 1,2,\cdots,N)$ indicates the real track point of the UAV. $y_i(i = 1,2,\cdots,N)$ shows the closest point on the corresponding contour line obtained after searching the sampling point of each inertial navigation system.

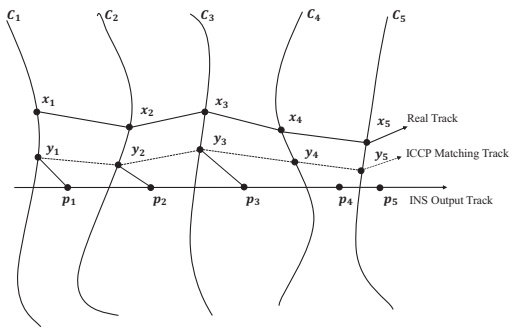

Fig.1: Basic Schematic Diagram of ICCP Algorithm

The ICCP algorithm assumes that the true trajectory point x_i must be located on a certain counter C_i, and the gravity anomaly value of the counter is equal to the gravity anomaly value measured at the corresponding inertial navigation system sampling point. In the background image of the gravity field after interpolation processing, the position point output by the inertial navigation system is projected onto the contour, then Use the Euclidean distance of formula (1) as the objective function to obtain the optimal estimated value.

$$M = \sum_{n=1}^{N} \| y_i - Rp_i - t \|^2 \tag{1}$$

In the formula above, R is the rotation matrix, t is the translation vector.

After determining the closest equivalent point, the ICCP algorithm uses the quaternion method to calculate the rigid transformation that satisfies the optimal objective function according to the principle of first rotation and then translation. Next the output point set of the inertial navigation system is transformed until the iterative termination condition is satisfied, and the optimal trajectory is used as the output to correct the inertial navigation system.

However, there are errors not only in the construction of the gravity field background map, but also in the gravimeter measurement. The source and feature of the error between the gravity field background map and the real-time gravity measurement are inconsistent.

3 Improved ICCP Algorithm Based on Mahalanobis Distance

Mahalanobis distance is one of the effective methods to study spatial vector correlation proposed by Indian statistician P. C. Mahalanobis. Based on the affine invariance of Mahalanobis distance which is independent of measurement scale, in this paper, Mahalanobis distance is used instead of Euclidean distance to improve the optimal objective function in the iterative process, and improve the matching accuracy of the ICCP algorithm, which makes the algorithm more practical.

Regard the inertial navigation output sequence $P = \{p_i(i = 1,2,\cdots,n)\}$ and the closest counter value set $Y = \{y_i(i = 1,2,\cdots,n)\}$ found in the gravity reference map as two sample sets respectively. When the traditional ICCP algorithm performs rigid transformation, the Euclidean distance optimal objective function is used in the iteration process. Figure 2 intercepts the closest point of the current contour search and the matching point after rigid iteration. It can be clearly seen that after a long period of navigation, due to the larger output error of the inertial navigation system, the initial matching position is not easy to determine, making the error of the gravity measurement and the gravity reference map cannot be ignored, which may lead to mismatch. The traditional ICCP algorithm does not comprehensively consider the influence of the gravity field background map error and the gravity real-time measurement error on the iterative process. In practice, the impact of the two on the matching algorithm will vary with the resolution of the gravity field background image and the accuracy of the gravity measuring instrument.

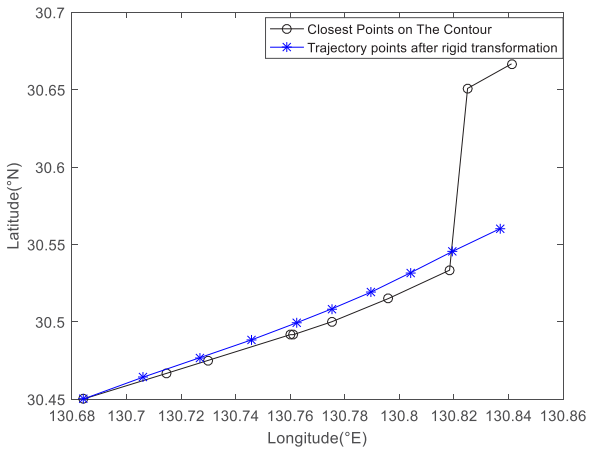

Fig.2: Trajectory Diagram of Points After Long-Time Sailing

The difference in the variance of the gravity field background map error and the gravity real-time measurement error makes the gravity data distributed in different forms around the average value in space. The data distribution characteristics of this asymmetric expansion can be expressed by the covariance structure. Mahalanobis distance is a kind of generalized distance that uses a covariance matrix to quantify the correlation between sample points and data sets, which makes it suitable for gravity matching. This paper uses Mahalanobis distance instead of Euclidean distance to establish a new objective function.

First, we use the gravity measurement data and the gravity reference map data to construct the matching error covariance matrix Σ. The covariance matrix Σ can de-correlate the gravity data measured by the output sequence of the inertial navigation system $P = \{p_i(i = 1,2,\cdots,n)\}$ with the data of the closest equivalent point set found in the gravity reference map $Y = \{y_i(i = 1,2,\cdots,n)\}$, so that the algorithm can perform rigid transformation more effectively.

Improve the Euclidean distance optimal objective function M to the Mahalanobis distance optimal objective function:

$$M = \sum_{i=1}^{n} (y_i - Rp_i - t)^T \Sigma^{-1} (y_i - Rp_i - t) \tag{2}$$

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on July 29,2024 at 06:58:43 UTC from IEEE Xplore. Restrictions apply.

In the formula above, R is the rotation matrix, t is the translation vector, $T = (R, t)$ is the rigid transformations.

The calculation of $(i, j)th$ term of the covariance matrix is given by equation (3):

$$\Sigma_{ij} = \mathrm{cov}(y_i, p_j) = E[(y_i - \bar{y})(p_j - \bar{p})] \tag{3}$$

In the formula above, $\bar{y}$、$\bar{p}$ are the center points of the samples.

The end of the iteration requires certain conditions. This article uses the following two formulas as the conditions for stopping the iteration.

$$\left|R^{(k+1)} - R^{(k)}\right| < \varepsilon_r, \left|t^{(k+1)} - t^{(k)}\right| < \varepsilon_t \tag{4}$$

Or $$k > k_{\max} \tag{5}$$

Among them, k means the number of iterations, formula (4) indicates that the rigid transformation $T = (R, t)$ result obtained in this iterative calculation has no significant change compared with the calculation result of the previous iteration, so the algorithm terminates the iteration at this time. Equation (5) is a termination condition to prevent the iteration from falling into an infinite loop, When the number of executions of the algorithm reaches the set upper limitk_{max}, the iterative process will be forcibly stopped. After iteration, a matching track that gradually moves closer to the real track can be obtained, which is used as the best matching track to correct the track output by the inertial navigation system.

4 Strip Area Searching Based on Measure Error and Objective Function

A key step in the ICCP algorithm is to search for the closest point on the contour map. However, when the traditional ICCP algorithm searches for the closest contour point corresponding to the sampling point of each inertial navigation system, the initial search field selected is usually a large circular area or even the entire gravity field background map. This approach will obviously increase unnecessary calculations. Therefore, this paper proposes a search strategy based on the strip area, which can effectively improve the real-time performance and computational efficiency of the ICCP algorithm. The key to the strip search lies in the selection of the normal width of the strip area and the length of the strip area along the counter direction.

4.1 Determination of Normal Width d of Strip Area

The normal width d of the strip area is the search distance in the normal direction of the contour with the current sampling point as the center. The basic idea to determine d is shown in Figure 3.

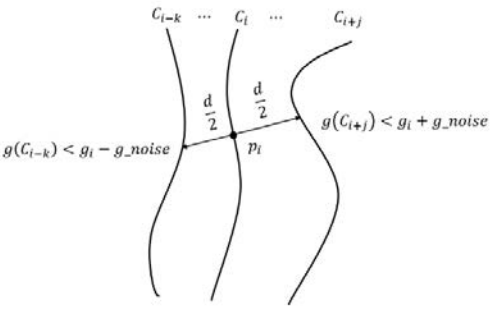

Fig.3: Schematic Diagram of Determining Width of Strip Area

In actual navigation measurement, the measurement error of the gravimeter is inevitable. Suppose the measurement noise error variance of the gravimeter is g_noise, The current sampling point is located on the contour line C_i, The actual gravity anomaly value measured by the gravimeter is g_i. The basic idea of determining the width d of the strip area is that based on the contour of the current sampling point, search along the normal direction of the contour until the gravity anomaly value on the contour meets one of the following conditions：

$$\begin{aligned} g(C_{i+j}) &\le g_i + g_noise \\ g(C_{i-k}) &\ge g_i - g_noise \end{aligned} \tag{6}$$

In the inequality above, $g(C_{i+j})$ is the gravity anomaly of counter C_{i+j}, $g(C_{i-k})$ is the gravity anomaly of counter C_{i-k}.

4.2 Determination of the Length of Strip Area

The specific expression form of the contour direction length of the strip area needs to be derived from the Mahalanobis distance objective function proposed above. Suppose the objective function of the ICCP algorithm is M, The minimal rigid transformation is $T = (R, t)$. The objective function of the iterative process is:

$$\begin{aligned} M &= \sum_{i=1}^{n} M_i = \sum_{i=1}^{n} (y_i - Rp_i - t)^T \Sigma^{-1} (y_i - Rp_i - t) \\ &= (y_1 - Rp_1 - t)^T \Sigma^{-1} (y_1 - Rp_1 - t) + (y_2 - Rp_2 - t)^T \Sigma^{-1} (y_2 - Rp_2 - t) + \cdots \\ &\quad + (y_n - Rp_n - t)^T \Sigma^{-1} (y_n - Rp_n - t) \end{aligned} \tag{7}$$

During the kth iteration, Calculating the minimum value of $M^{(k)}$ is equivalent to calculating the minimum value of the objective function of each inertial navigation system INS sampling point $M_i^{(k)} = (T^{(k)} p_i^{(k)} - y_i^{(k)})^T \Sigma^{-1} (T^{(k)} p_i^{(k)} - y_i^{(k)})$.

In order to make the iteration meaningful, each rigid transformation of the point set P,must make the matching trajectory closer to the real trajectory, that is, satisfy the following inequality:

$$M_i^{(k)} \le e_i^{(k)} \tag{8}$$

Let the residual sum of squares after matching be $e_i^{(k)} = ||p_i^{(k)} - y_i^{(k)}||^2$, in which i indicates the ith sampling point of the inertial navigation system. At the same time, substituting equation (7) into inequality (8), we get:

$$|(T^{(k)} p_i^{(k)} - y_i^{(k)})^T \Sigma^{-1} (T^{(k)} p_i^{(k)} - y_i^{(k)}) \le \| p_i^{(k)} - y_i^{(k)} \|^2 \tag{9}$$

Assuming that the above inequality is not satisfied, then the equivalent of the rigid body transformation of the point set P will produce a larger mean square error than the original point set, which means that the iteration is meaningless, which is obviously impossible to happen.

According to the basic process of iteration:

$$p_i^{(k+1)} = T^{(k)} p_i^{(k)} \tag{10}$$

Substitute optimal objective function $M^{(k)} = (T^{(k)} p_i^{(k)} - y_i^{(k)})^T \Sigma^{-1} (T^{(k)} p_i^{(k)} - y_i^{(k)})$：

$$M_i^{(k)} = (p_i^{(k+1)} - y_i^{(k)})^T \Sigma^{-1} (p_i^{(k+1)} - y_i^{(k)}) \tag{11}$$

And the error between two iteration matches is set to $e_i^{(k+1)} = ||p_i^{(k+1)} - y_i^{(k+1)}||^2$. Because $y_i^{(k+1)}$ is the closest point on the contour found according to the new round of sampling sequence $p_i^{(k+1)}$, so we can get：

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on July 29,2024 at 06:58:43 UTC from IEEE Xplore. Restrictions apply.

$$e_i^{(k+1)} \le M_i^{(k)} \tag{12}$$

According to formula (12), let $k = k + 1$:

$$M_i^{(k+1)} \le e_i^{(k+1)} \tag{13}$$

In summary, combining formulas (11), (12), (13), it can be obtained that:

$$M_i^{(k+1)} \le e_i^{(k+1)} \le M_i^{(k)} \le e_i^{(k)} \tag{14}$$

since that $M_i^{(k+1)} \le M_i^{(k)}$, and then:

$$\| T^{(k+1)} p_i^{(k+1)} - y_i^{(k+1)} \| \le M_i^{(k)} \tag{15}$$

According to equation (15), the search length of the strip area along the contour can be obtained:

$$l_{k+1} \le M_i^{(k)} \tag{16}$$

That is, the length of the $(k+1)th$ strip search does not exceed the objective function value of the kth iteration. The above is the conclusion drawn by the Mahalanobis distance derivation, and it can be verified that the conclusion is still valid under the Euclidean distance.

After the derivation above, we can get that at the kth iteration, the closest points on the contour map which is corresponding to the sampling points of each inertial navigation system will be distributed in the strip area with the sampling point $p_i^{(k+1)}$ as the center while the width is d and length is l.The width and length are determined by formula (6) and formula (16) respectively.

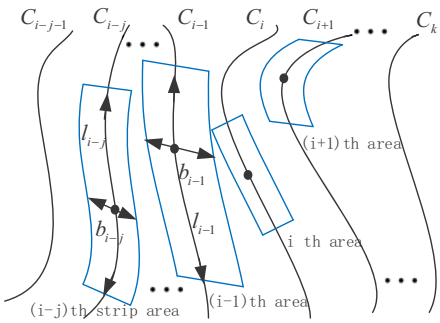

Fig4: Schematic Diagram of Strip Search Area

It can be seen from Figure 4 that after each iteration, the search area of the strip area will be reduced, which is also related to the convergence of the ICCP algorithm. With the gradual increase of the number of iterations, the area of the strip area will become smaller, so that the calculation amount of the algorithm when finding the closest point on the contour is gradually reduced. Moreover, compared to other methods that use circular or square areas to narrow the search area, the strip search area takes the unevenness of the contours in the gravity anomaly map to a certain extent, the search area will be curved along the curve of the contour, which increases the possibility of searching for the global optimum.

5 Simulation and Analysis

The improved ICCP matching algorithm proposed in paper is shown in Figure 5.

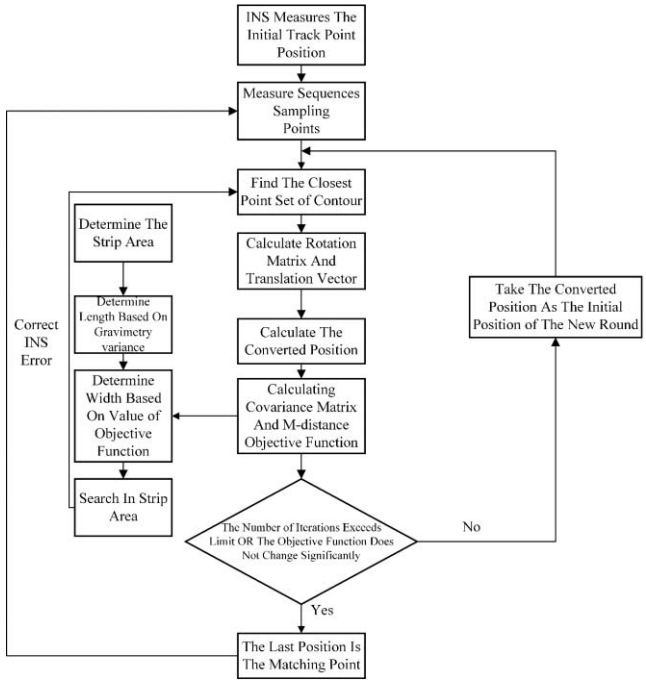

Fig.5: Flow Chart of Improved ICCP Algorithm

All of simulations in this paper are constructed based on MATLAB. The accuracy of the gravity anomaly background image used in the simulation experiment is,$1.5' \times 1.5'$, the longitude range is 130°E~ 131°E, The latitude range is 30°N~ 31°N, The variation range of gravity anomaly is -40.3808mGal~69.8142mGal. Set the starting point of the real track at 130.2°E、30°N. The underwater vehicle makes a straight-line sailing at a speed of $10nmile/h$ in a direction of 45 east by north, the simulation time is 280min. Add a Gaussian white noise with a mean value of 0.6mGal and a variance of 0.1mGal to the real-time measurement data of the gravimeter as a real-time measurement sequence of gravity anomalies. The track of SINS is obtained by simulation. Figure 6 is the matching trajectory calculated by the traditional ICCP algorithm and the improved ICCP algorithm in this paper. Figure 7 is a comparison figure of latitude and longitude position errors between the improved ICCP algorithm and the traditional ICCP algorithm. Figure 8 is a comparison figure of the running time of the improved ICCP algorithm and the traditional ICCP algorithm.

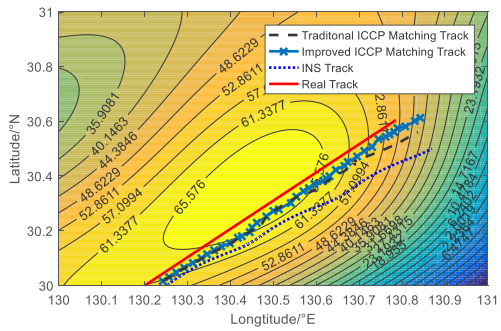

Fig.6: Improved ICCP Algorithm Simulation

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on July 29,2024 at 06:58:43 UTC from IEEE Xplore. Restrictions apply.

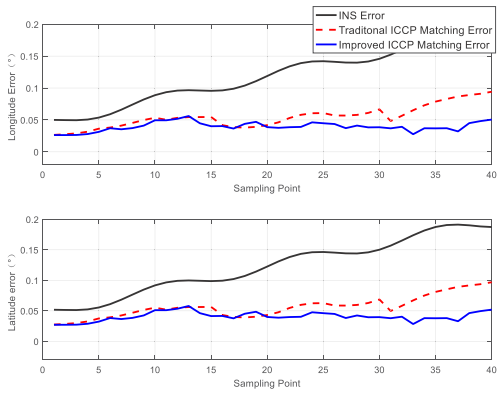

Fig.7: Error Result Graph of Improved ICCP Algorithm

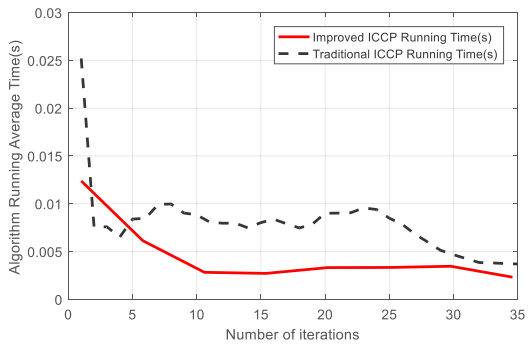

Fig.8: Running Time of Improved ICCP Algorithm

It can be seen from Figure 7 and Figure 8 that compared with the traditional ICCP algorithm, after the improvement of Mahalanobis distance objective function and the strip search area, the improved ICCP algorithm has better matching positioning accuracy and real-time performance. Especially after a long voyage, the convergence of the improved algorithm is stronger. The error will not diverge, and the running time of the algorithm is very short.

Table 1: Error Statistics of Improved ICCP Algorithm

Navigation method		SINS	Traditional ICCP	Improved ICCP
Lat. Error / °	Avg.	0.12329	0.056434	0.041701
	S.D.	0.045368	0.018057	0.0074704
Lon. Error / °	Avg.	0.11961	0.054731	0.039559
	S.D.	0.044211	0.017595	0.0072518
Pos. Error / °	Avg.	0.17177	0.078568	0.057438
	S.D.	0.062936	0.025104	0.010409

From the Table 1, we can conclude that compared with traditional ICCP algorithm, the mean position error of the improved ICCP algorithm is reduced by 26.89%, and the standard deviation is reduced by 58.53%. At the same time, the mean latitude error is reduced by 26.11%, and the mean longitude error is reduced by 27.72%. The average running time of the improved algorithm is also 42.1% shorter than the traditional ICCP algorithm. The simulation results show that the improved ICCP algorithm not only runs less than the traditional algorithm, but also has strong real-time performance, and improves the matching positioning accuracy.

6 Conclusion

To solve the limitation of accuracy and real-time performance in traditional ICCP algorithm, this paper proposes an improved ICCP gravity matching algorithm based on Mahalanobis distance. At the same time, a new ICCP algorithm searching strategy based on the strip search area is proposed to narrow the search range of the closest contour point in the algorithm. This improves the real-time performance of the matching algorithm and better meets the needs of underwater vehicles. After simulation experiments, the improved ICCP algorithm has clearly higher matching positioning accuracy. In summary, the improved ICCP gravity matching algorithm based on Mahalanobis distance reduces the running time of the algorithm, improves real-time performance and positioning accuracy, and further improves the practicability of gravity matching and positioning for underwater vehicles.

References

[1] Fu Mengyin, Liu Fei, Yuan Shuming. Overview of underwater inertial/gravity matching autonomous navigation[J]. *Journal of Underwater Unmanned Systems*, 2017, 25(1): 31-43.

[2] Wang Bo, Zhou Minglong. Research progress on the selection of underwater gravity assisted navigation adaptation area [J]. *Journal of navigation and positioning*, 2020,8 (03): 32-39.

[3] Wang Bo, Fu Mengyin, Li Xiaoping, Wu Taiqi, Deng Zhihong. Overview of underwater gravity matching positioning algorithm [J]. Navigation and control, 2020,19 (z1): 170-178.

[4] Yuan Shuming, Sun Feng, Liu Guangjun, Chen Jing. Application of gravity figure matching technology in underwater navigation[J]. *Journal of Chinese Inertial Technology*, 2004, 12(2): 13-17.

[5] Deng Zhihong, Fu Mengyin, Zhang Jiwei, et al. Inertial devices and inertial navigation systems [M]. Beijing: Science Press, 2012.

[6] Yan Li, Cui Chenfeng, Xie Hong. The application of ICCP algorithm in gravity matching[J]. *Remote Sensing Information*, 2009, 1:16-19.

[7] Zheng Tong, Bian Shaofeng, Wang Zhigang. Submarine terrain matching aided navigation based on ICCP matching algorithm[J]. *Ocean Surveying and Mapping*, 2008, 28(2): 21-23.

[8] Yu Li. Research on Gravity Matching Algorithm of Underwater Carrier[D]. Beijing: Beijing Institute of Technology, 2016.

[9] Moryl J, Rice H, Shinners S. The Universal Gravity Module for Enhanced Submarine Navigation[C]. *Position Location and Navigation Symposium, IEEE. IEEE*, 2002:324-331.

[10] Bai Wenping, Wang Zhigang. Research and improvement of ICCP gravity matching aided navigation algorithm[J]. Computer Simulation, 2013, 30(6): 15-19.

[11] Jiaolong Yang, Hongdong Li, Dylan Campbell, Yunde Jia. Go-ICP: A Globally Optimal Solution to 3D ICP Point-Set Registration[C]. *Transactions on Pattern Analysis and Machine Intelligence. IEEE. IEEE*, 2016:2241-2254.

[12] Liu Chengxiang. Research on terrain matching assisted positioning technology for underwater vehicles [D]. Harbin: Harbin Engineering University, 2003.

[13] Han Yurong. Research on gravity matching algorithm for underwater navigation [D]. Beijing: Beijing Institute of Technology, 2017.

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on July 29,2024 at 06:58:43 UTC from IEEE Xplore. Restrictions apply.

[14] Zheng Tong. Research on Submarine Terrain Matching Aided Navigation Technology [D]. Wuhan: Naval University of Engineering, 2008.

[15] Madden Paul . Passive Autonomous Navigation For Exploration, Defense Using Gravity to Aid INS on UUVs[J]. *Sea Technology,* 2017，58(6)：37-40.

[16] Dong Cuijun. Research on the matching algorithm and the selection of adaptive area for underwater gravity field assisted inertial navigation [d]. Wuhan: Wuhan University, 2017

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on July 29,2024 at 06:58:43 UTC from IEEE Xplore. Restrictions apply.